

Asset Pricing in the 21st Century: A Critical Evaluation of Traditional Models **vs. Market Complexity**

UNH (UnitedHealth Group) — 20 Years of Daily Data (2004–2024)

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Agenda

What we cover in this presentation

01

The Debate

Cochrane's (2009) parsimonious SDF framework vs. the complexity view of modern markets. Can traditional equilibrium models explain today's pricing anomalies?

02

Stock Analysis — UNH

Normality testing via Jarque-Bera, correlation analysis with S&P 500 and NASDAQ benchmarks, and descriptive statistics of UNH daily returns.

03

Random Walk Model

ADF stationarity tests on price levels vs. log returns, followed by Monte Carlo simulation to model potential future price paths under random walk assumptions.

04

CAPM Beta Analysis

OLS regression-derived beta estimation, comparison with CAPM-predicted beta, and testing Jensen's Alpha for statistical significance of abnormal returns.

05

Conclusion

A hybrid approach: CAPM provides a robust equilibrium baseline, while complexity tools capture tail risks, regime shifts, and non-linear dynamics that traditional models miss.

The Debate: Cochrane vs. Complexity View

Two opposing frameworks for understanding modern asset prices

Cochrane (2009) — Traditional View

- Core Equation:** $p = E(mx)$ — price equals expected discounted payoff
- SDF Framework:** unifies CAPM, ICAPM, and consumption-based models
- Factor Models:** Fama-French 3-factor (HML, SMB) explains cross-sectional returns
- Rational Expectations:** investors maximize utility; markets are informationally efficient
- Parsimony:** mathematically elegant with few free parameters
- Equity Premium Puzzle:** utility functions can't reconcile the ~6% historical premium
- Volatility Puzzle:** consumption too smooth to justify observed return volatility

VS

Complexity View — Modern Markets

- Bounded Rationality:** heuristics and cognitive biases drive systematic mispricing
- Non-Linear Dynamics:** small shocks amplify through interconnected networks
- Fat Tails:** excess kurtosis and skewness beyond normal assumptions
- Algorithmic Feedback:** HFT and systematic strategies create liquidity spirals
- Agent-Based Models:** heterogeneous agents with adaptive strategies
- Crash Explanation:** better accounts for contagion and cascading failures
- Regime Shifts:** structural breaks challenge single-model frameworks

Part 1: Normality of UNH Daily Returns

Jarque-Bera test, histogram vs. normal fit, Q-Q plot | 5,283 observations (2004–2024)

0.9607

Skewness

31.19

Excess Kurtosis

214,478

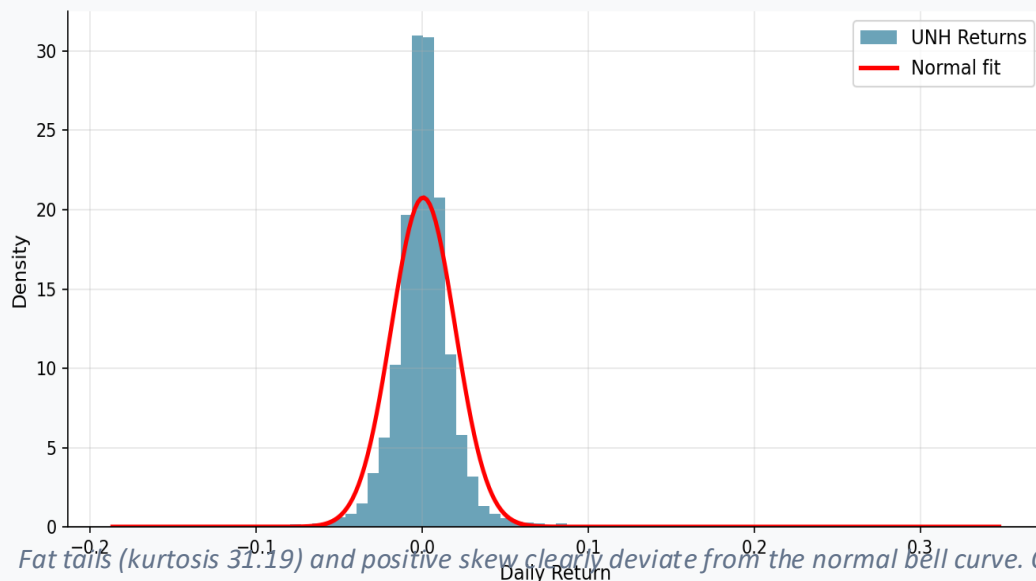
Jarque-Bera Stat

$p \approx 0.000$

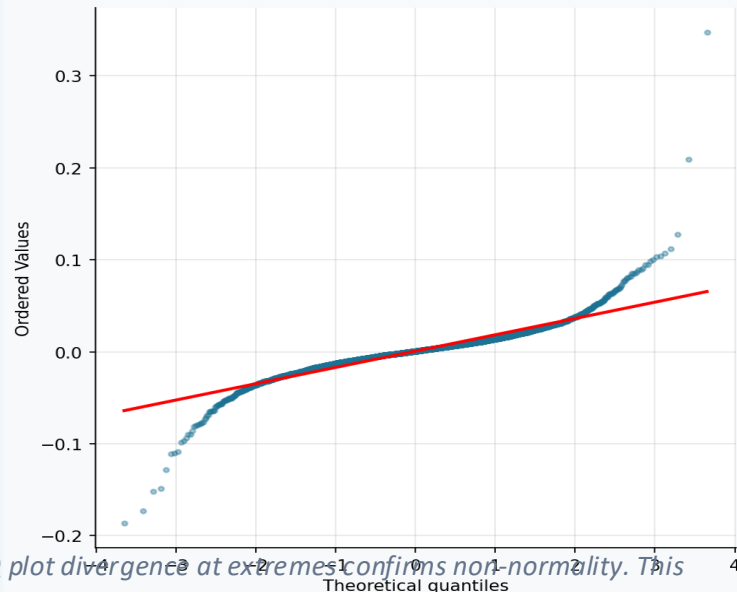
JB p-value

**REJECT
 H_0**

UNH Daily Return Distribution



Q-Q Plot of UNH Returns vs Normal

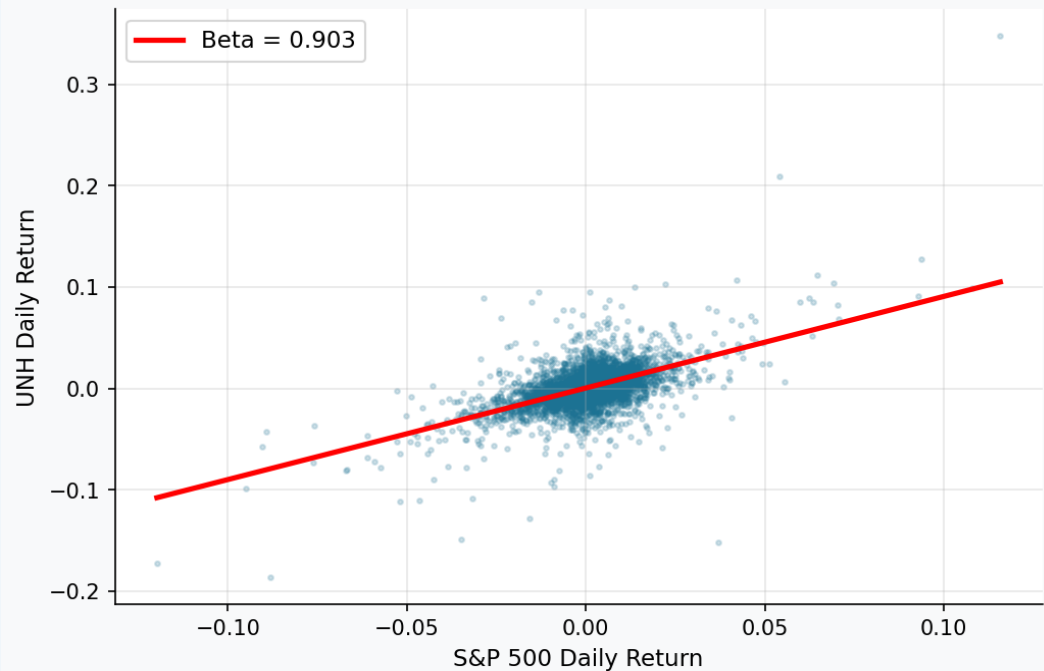


Fat tails (kurtosis 31.19) and positive skew clearly deviate from the normal bell curve. Q-Q plot divergence at extremes confirms non-normality. This challenges CAPM's Gaussian assumption.

Part 1: Relationship with the Market

OLS regression of UNH returns against S&P 500 and NASDAQ | 20-year daily data

UNH vs S&P 500 Returns | Beta = 0.903



Metric	UNH	S&P 500	NASDAQ
Mean Daily Return	0.0765%	0.0388%	0.0521%
Std Deviation	1.9212%	1.1885%	1.3467%
Min Return	-18.64%	-11.98%	-12.32%
Max Return	+34.76%	+11.58%	+11.81%
Observations	5,283	5,283	5,283

Regression Summary

Metric	vs S&P 500	vs NASDAQ
Correlation r	0.5588	0.4887
R-squared	0.3123	0.2388
Alpha (p)	0.000415 (0.059)	0.000402 (0.082)
Beta	0.9034 ***	0.6971 ***

UNH beta of 0.90 suggests near-market sensitivity. Healthcare’s inelastic demand keeps UNH slightly less volatile than the broad market. Alpha is statistically insignificant — no excess return beyond systematic risk.

Part 2: Random Walk & Time-Series Modelling

Augmented Dickey-Fuller stationarity test + Monte Carlo simulation (100 paths, 1 year)

-0.0119

ADF stat (log prices)

0.9575

p-value (log prices)

-16.616

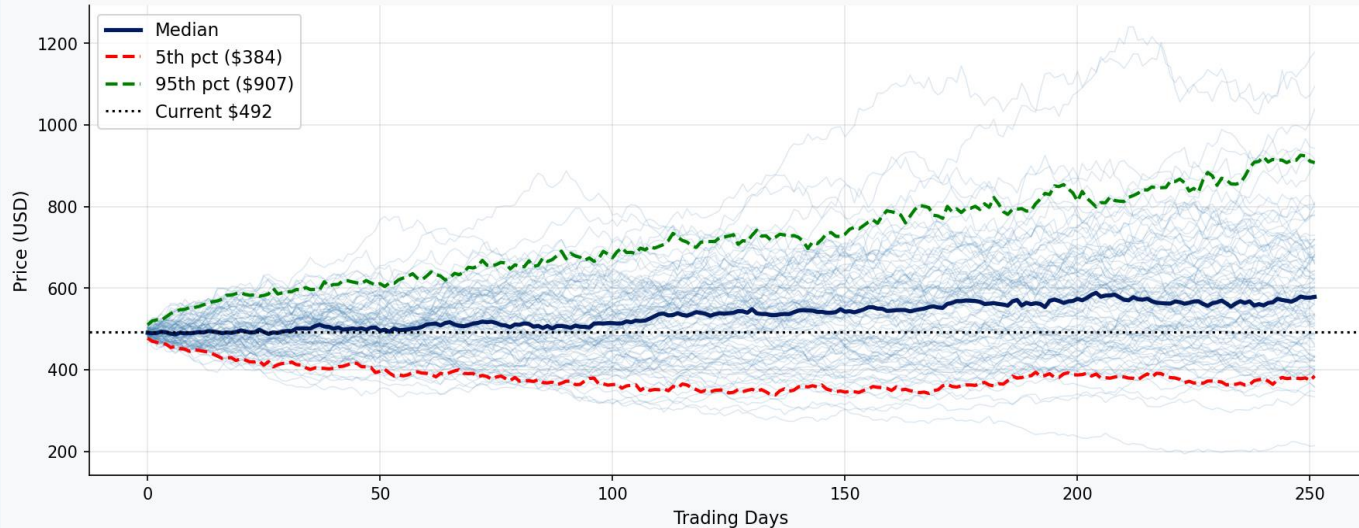
ADF stat (returns)

$p \approx 0.000$

p-value (returns)

Log prices: ADF stat > critical value (-2.86) $\Rightarrow$ Cannot reject unit root $\Rightarrow$ UNH log-prices follow a Random Walk. | Returns: stationary ($p \approx 0$), confirming the random walk in price levels.

UNH Monte Carlo Simulation (100 paths, 1 year)



\$491.77

Starting Price

\$578.34

Median 1yr

\$384.47

5th pct

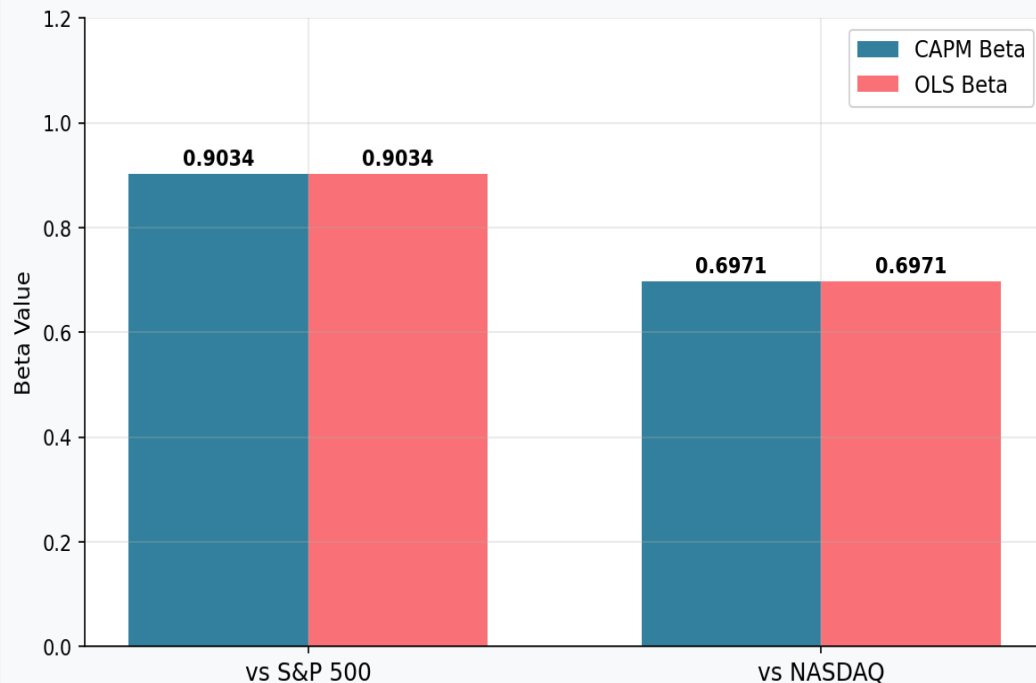
\$907.33

95th pct

Part 3: CAPM Beta vs. Observed OLS Beta

Risk-free rate: 4.3% p.a. (US 10-yr Treasury = 0.01706% daily) | Excess return regressions

CAPM Beta vs OLS Beta - UNH



CAPM Analysis Summary

Metric	vs S&P 500	vs NASDAQ
CAPM Beta	0.9034	0.6971
OLS Beta	0.9034	0.6971
Difference	0.000000	0.000000
Jensen Alpha	0.000398	0.000350
Alpha p-value	0.069 (n.s.)	0.129 (n.s.)

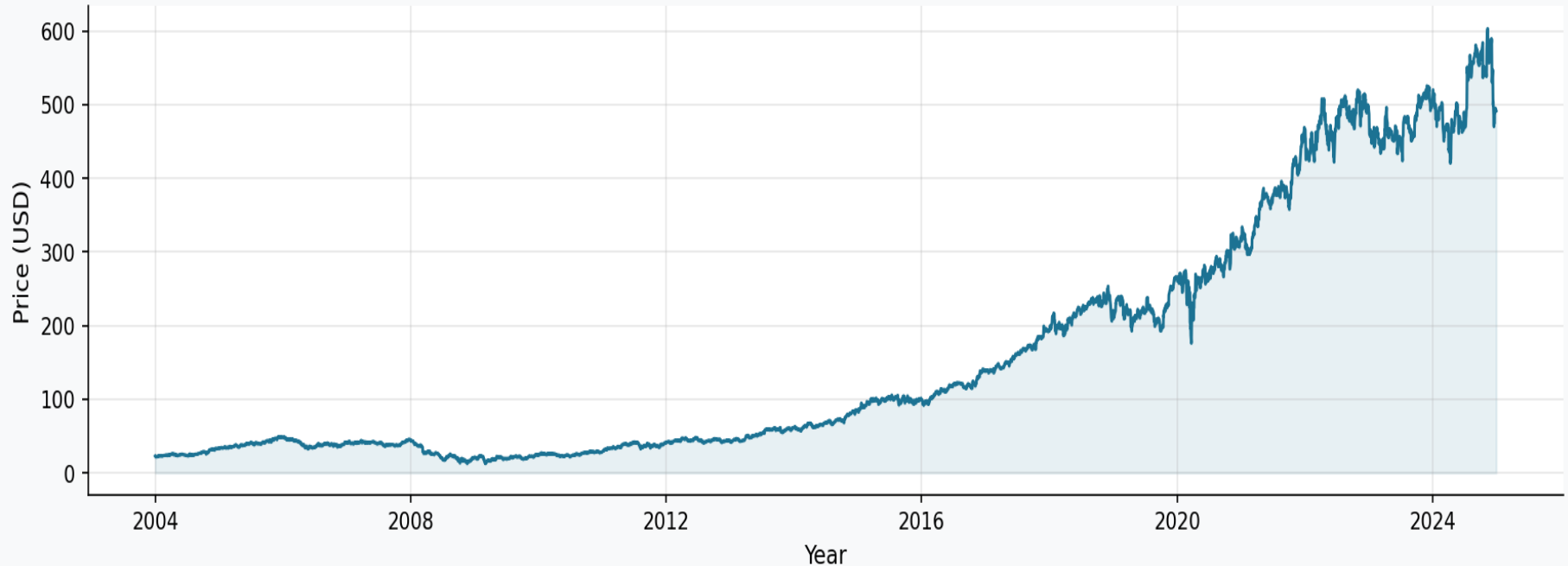
CAPM and OLS betas are virtually identical. Jensen's Alpha is not statistically significant vs. S&P 500 — UNH earns no excess return beyond what CAPM predicts. Beta < 1 reflects defensive healthcare characteristics.

$R_f \text{ daily} = 4.3\% / 252 = 0.01706\%$ | $UNH \text{ Beta (vs S\&P 500)} = 0.9034$, suggesting near-market risk exposure

UNH 20-Year Price History (2004–2024)

From \$22.76 to \$491.77 — a 21x price appreciation over the analysis window

UNH Closing Price (2004-2024)



2008–09
GFC

2020
COVID Crash

2022
Rate Hikes

Conclusion

What the evidence tells us about UNH and asset pricing models

For Cochrane

- Beta stable and significant (~ 0.90)
- CAPM correctly identifies systematic risk
- Jensen Alpha insignificant vs S&P 500
- SDF framework holds for average returns
- No evidence of persistent alpha generation

For Complexity

- Non-normal returns (kurtosis = 31.19)
- Jarque-Bera $p < 0.00001$ rejects normality
- \$523 simulation range (5th-95th pct)
- CAPM underestimates tail risk
- Fat tails persist across 20-year window

Our View — Hybrid

- CAPM for equilibrium pricing baseline
- Monte Carlo for forward-looking scenarios
- Tail-risk metrics for stress periods
- Complexity tools supplement, not replace
- Neither model alone is sufficient

Group 9 — Members

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